Sean Luka Girgis

Senior Hadoop / Production Support Data Engineer

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LinkedIn | GitHub | Portfolio

Professional Summary

Senior data engineering and production support professional with 20+ years of enterprise IT experience across financial services, telecom, travel, and large-scale infrastructure environments. Strong background in Linux/Unix troubleshooting, SQL/PL/SQL, Hadoop/Hive/Spark, ETL/data pipeline support, data validation, incident analysis, RCA, runbooks, and stakeholder communication. Experienced supporting high-availability production platforms, analyzing logs and telemetry, resolving data issues, and improving operational reporting and automation.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Supported enterprise-scale data and infrastructure workflows in a regulated financial-services environment, combining Python, SQL, PySpark, Hadoop/Hive, and Linux/Unix troubleshooting to maintain reliable reporting and capacity analytics pipelines.
- Built and maintained automated ETL/data pipelines that ingested, cleaned, normalized, and validated telemetry from thousands of infrastructure endpoints for downstream forecasting, reporting, and operational decision support.
- Investigated pipeline, data quality, and reporting issues by tracing upstream/downstream dependencies, reviewing logs and source data, validating SQL outputs, and coordinating with application, database, infrastructure, and stakeholder teams.
- Used SQL and Oracle-based data analysis to reconcile data, investigate anomalies, support historical retention, and validate operational metrics across enterprise monitoring and capacity datasets.
- Partnered on scaling data processing patterns from Pandas-based workflows toward PySpark, Hadoop, and cloud-scale processing approaches for larger telemetry and forecasting workloads.
- Created operational reports, dashboards, risk rankings, and runbook-style documentation that helped teams prioritize remediation, capacity planning, resource allocation, and production stability actions.
- Improved monitoring, alerting, and automation around capacity and telemetry pipelines, helping surface exceptions, bottlenecks, and threshold risks before they became operational incidents.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed production performance monitoring for Brand.com and critical hospitality systems using Dynatrace AppMon, Gomez Synthetic Monitoring, Linux, Oracle, PostgreSQL, and data analytics workflows.
- Analyzed synthetic and real-user performance data to identify system issues, transaction bottlenecks, and optimization opportunities for engineering and operations teams.
- Performed production monitoring upgrades and TLS-related cutover work with careful validation to avoid monitoring regression during the transition.
- Integrated load-test and APM visibility so teams could correlate performance issues, logs, transactions, and infrastructure behavior during test and production readiness cycles.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Supported large-scale enterprise monitoring environments for a financial-services client, including 50+ Enterprise Managers and thousands of instrumented agents across critical applications.
- Used Linux/Unix, Oracle, SQL, Perl, and Korn shell scripting to automate performance reporting, extract operational data, and reduce manual support work.
- Led troubleshooting, upgrades, dashboards, alert thresholds, SLA views, and technical documentation for high-availability monitoring environments.
- Worked closely with J2EE, WebLogic, WebSphere, database, application, and operations teams to diagnose performance issues and support production stability.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration and validation work for a high-throughput transaction platform moving from a large MySQL estate to Oracle RAC, emphasizing data correctness, performance, and operational continuity.
- Built C++/OCCI/OCI and SQL-based validation and testing workflows to verify migration correctness across large transaction datasets.
- Troubleshot database, application, and performance issues across Linux/Unix-style enterprise environments while supporting a critical travel transaction platform.

Developer / Support Engineer at Corpus Inc. / Sprint (AMDOCS projects) (2001 – 2007)

- Developed and supported telecom billing, reporting, and interface processes using C, C++, Pro*C, PL/SQL, Oracle, Korn shell, Perl, XML, and Unix platforms.
- Performed production bug triage, data validation, troubleshooting, and batch/interface support across billing, settlement, reporting, and downstream integration workflows.
- Automated WebLogic/WebSphere administration and operational support tasks using Korn shell scripting, reducing manual deployment and support steps.
- Maintained code and support processes across HPUX, SunOS, and Linux environments while collaborating with QA, development, and production support teams.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt / Cairo, Egypt

Skills

Flagship Projects

HorizonScale — Capacity Forecasting and Telemetry Pipeline
Built and evolved a capacity forecasting and telemetry analytics workflow that transformed raw operational metrics into validated features, forecasts, threshold risk flags, risk rankings, and stakeholder reports. Started with SQL/Python/Pandas analysis and moved toward PySpark/Hadoop/cloud-scale patterns for larger data volumes.
Technologies: Python, SQL, Pandas, PySpark, Hadoop/Hive, Prophet, scikit-learn, Streamlit, Linux/Unix

Enterprise Monitoring and APM Reporting Automation
Automated operational reporting and monitoring analysis across enterprise APM and telemetry platforms, using scripting, SQL, dashboards, alerts, and documentation to support incident investigation, performance analysis, and production readiness.
Technologies: CA APM, Dynatrace, BMC TrueSight, AppDynamics, SQL, Perl, Korn Shell, Linux/Unix

AWS / Serverless Data Platform Study and Implementation
Designed and practiced cloud data platform patterns using S3, Glue, Athena, Redshift, Parquet, and PySpark, mapping prior enterprise ETL and data warehousing experience to modern cloud data engineering approaches.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet, Python